

AAKASH SUDANAGUNTA

Business Data Analyst

Email:

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PROFESSIONAL SUMMARY:

3 years of progressive experience as a Business/Data Analyst across financial services, healthcare, and IT product domains; designed analytics solutions that enhanced reporting efficiency, portfolio planning, and clinical resource management.

Proficient in SQL, Python, R, and Power BI/Tableau for data integration, cleaning, modeling, and visualization; optimized SQL queries and database indexing to improve retrieval performance by up to 30% in financial data workflows.

Developed and automated 20+ ETL pipelines using tools like MySQL, BigQuery, and AWS Glue, streamlining ingestion and transformation of high-volume datasets for credit product analytics and healthcare insights.

Designed interactive dashboards using DAX, Power BI, and Tableau to deliver KPI-driven insights for product performance, card member engagement, and patient treatment outcomes, enabling real-time strategic decisions.

Built machine learning models including patient risk segmentation (85% accuracy) and admission forecasting (92% accuracy) using Python (Pandas, Scikit-learn), enhancing care delivery and operational planning in hospital networks.

Engineered and documented dimensional data models (Star Schema) and data warehouses to support structured analytical querying and scalable reporting for business stakeholders across finance and healthcare operations.

Conducted advanced statistical analyses including hypothesis testing, logistic regression, and scenario modeling to evaluate credit allocation strategies and identify sales anomalies, improving financial reporting accuracy and marketing impact.

Collaborated with product, compliance, and engineering teams to define and refine business requirements; authored quarterly analytics packages and presentations for audit readiness and internal performance reviews.

Enhanced reporting operations by building automation scripts in Excel using VBA macros, INDEX-MATCH, and Power Query to validate and reconcile large-scale datasets for finance and sales metrics.

Led cross-functional efforts to identify process inefficiencies and improve data accuracy by implementing validation frameworks and optimizing data workflows across SQL databases and BI tools.

Delivered patient-centric dashboards and sales performance trackers for executive stakeholders, combining technical precision with clear visual storytelling to drive tactical decisions.

Applied robust data validation and profiling techniques to ensure consistency and quality of incoming data from multi-source systems in compliance with regulatory and operational standards.

TECHNICAL KNOWLEDGE:

PROFESSIONAL EXPERIENCE:

Client: American Express, USA (Remote) July, 2024 – Present

Role: Data Analyst

Project Summary:

Key contributor to enterprise-level credit analytics initiatives aimed at enhancing data-driven decision-making and reporting efficiency across American Express's consumer credit portfolio.

Collaborated with cross-functional teams to modernize legacy financial reporting systems using cloud-based ETL solutions, interactive dashboards, and scenario modeling to support credit allocation, audit compliance, and business performance monitoring.

Functional Responsibilities:

Collaborated with product, compliance, and engineering teams to gather and document analytics and reporting requirements, translating them into structured BRDs and FRDs using Confluence and SharePoint for stakeholder alignment and audit readiness.

Designed and maintained 20+ automated ETL pipelines using SQL and Python, integrating financial data from transaction systems into centralized repositories to support scalable analytics and regulatory compliance.

Developed and optimized SQL queries and data models to enhance query performance and enable timely access to real-time credit, transaction, and risk data; implemented indexing and normalization techniques resulting in a 30% improvement in processing speed.

Built and deployed interactive Power BI and Tableau dashboards to provide executive and operational stakeholders with real-time visibility into key metrics such as credit usage trends, delinquency rates, charge-offs, and portfolio segmentation.

Applied advanced analytics techniques including scenario modeling and trend forecasting to evaluate credit resource allocation strategies, directly influencing planning decisions across portfolios exceeding \$100M in exposure.

Created and delivered quarterly analytics packages used for internal audits, regulatory reporting, and board-level presentations; automated manual reporting workflows to reduce turnaround time by 30%.

Conducted data quality audits and validation checks using advanced Excel tools (INDEX-MATCH, VLOOKUP, macros), ensuring the accuracy of financial and operational datasets used in executive reporting and compliance submissions.

Spearheaded the development of data visualizations that integrated KPIs across risk, operations, and finance teams, enabling data-driven interventions in credit policy and operational risk areas.

Supported business and technology alignment initiatives by documenting data lineage, operational workflows, and transformation logic across credit decision systems and reporting environments.

Partnered with business leaders to conduct gap analyses and uncover data-driven opportunities for improving reporting granularity and customer segmentation accuracy.

Participated in Agile ceremonies using JIRA and Azure DevOps, driving backlog grooming, sprint planning, and cross-functional delivery of data analytics features and reporting enhancements.

Facilitated training sessions for business teams on interpreting dashboard insights, understanding forecast outputs, and utilizing self-service BI tools for ad-hoc credit risk analysis.

Tech Stack: Power BI, Tableau, SQL Server, Python (Pandas, NumPy), MySQL, BigQuery, AWS Glue, Redshift, Excel (VBA, Macros), Confluence, SharePoint, JIRA, Azure DevOps, Financial Transaction Systems, KPI Dashboards, Scenario Modeling, Data Warehousing (Star Schema), ETL Automation, Credit Risk Analytics.

Client: OSS Technologies, Bangalore, INDIA August 2021 – July 2022

Role: Business Analyst

Project Summary:

Worked as a Business Analyst supporting data-driven transformation initiatives in the healthcare technology division of OSS Technologies. Focused on enhancing visibility into clinical service utilization, patient demographics, and operational KPIs by developing predictive models, performance dashboards, and structured data workflows to support hospital operations, quality of care analysis, and cost optimization.

Functional Responsibilities:

Collaborated with hospital administration, patient services, and clinical analytics teams to gather reporting requirements for service utilization, care delivery performance, and operational efficiency metrics.

Designed and developed interactive Power BI dashboards to track patient admissions, departmental workload, procedure volumes, and physician performance metrics—enabling improved resource planning and care quality oversight.

Extracted, cleaned, and transformed 30,000+ patient and treatment records from MySQL databases to support operational reporting, population health segmentation, and clinical service analysis.

Conducted patient segmentation using statistical modeling techniques to identify high-risk cohorts, analyze readmission trends, and inform care pathway adjustments across specialty services.

Utilized Excel tools (PivotTables, INDEX-MATCH, VLOOKUP, and Macros) to build robust validation templates for reconciling billing records, clinical events, and electronic claims data across systems.

Performed statistical hypothesis testing (t-tests, Chi-square tests) to identify significant differences in patient outcomes across treatment groups, supporting evidence-based clinical decision-making. Created structured reporting templates and ad-hoc reports to support hospital leadership during quality improvement initiatives, compliance audits, and budget planning cycles.

Collaborated with IT teams and departmental heads to define business rules and workflows for ETL processes, ensuring accurate integration of clinical and financial data for downstream analytics.

Documented data flows, transformation logic, and visual reporting standards in Confluence and SharePoint to support data governance and interdepartmental alignment.

Delivered data-backed recommendations to improve outpatient service distribution, reduce patient wait times, and optimize clinical throughput based on trend and correlation analysis.

Participated in Agile-based sprint planning to prioritize analytics deliverables and enhance communication between data engineering, compliance, and clinical stakeholders.

Tech Stack: Power BI (DAX, Power Query), Excel (VBA, INDEX-MATCH, Macros, PivotTables), MySQL, Python (Pandas, NumPy), Healthcare Operations Reporting, Patient Segmentation, Statistical Testing (t-tests, Chi-square), ETL Design, Confluence, SharePoint, Clinical KPI Dashboards, Care Quality Analytics, HIPAA-aligned Workflow Documentation, Agile Methodology.

Client: Robocoupler Pvt Ltd, Visakhapatnam, INDIA February 2021 – August 2021

Role: Data Analyst Intern

Project Summary:

Contributed to the development and automation of sales performance analytics and dashboard reporting for a mid-sized retail analytics platform. Supported data-driven sales optimization strategies by developing interactive BI dashboards, enhancing data quality controls, and performing segmentation analysis to improve stakeholder decision-making and customer insight.

Functional Responsibilities:

Collaborated with sales and marketing teams to define business requirements and key performance indicators (KPIs) for product-level and regional sales reporting dashboards.

Developed interactive Power BI dashboards using DAX, Power Query, and Excel, visualizing sales metrics such as revenue growth, profit margins, order trends, and return rates to support strategic performance tracking.

Conducted customer segmentation analysis using demographic and transactional data to identify high-value customer cohorts, enabling targeted promotions and improving conversion rates.

Implemented robust data validation processes using Python (Pandas) and Excel formulas (VLOOKUP, INDEX-MATCH) to enhance data accuracy and consistency across CRM-sourced datasets and sales records.

Designed custom Power BI visual templates to standardize reporting visuals, improving report clarity and increasing engagement from internal business users and sales executives.

Performed exploratory data analysis (EDA) on sales datasets to uncover performance anomalies, regional revenue gaps, and customer churn signals.

Automated data refresh pipelines for recurring dashboards, reducing manual workload and ensuring timely updates for monthly and quarterly business reviews.

Created Excel-based sales reporting templates and validation sheets to support ad hoc and static reporting workflows used in executive briefings and sales reviews.

Documented BI report logic, data source mapping, and transformation steps in a structured format to support transparency, knowledge transfer, and future scalability.

Participated in Agile-based project cycles, supporting iterative delivery of dashboard enhancements and aligning development sprints with evolving sales analytics needs.

Tech Stack: Power BI, DAX, Power Query, Excel (PivotTables, VLOOKUP, Macros, INDEX-MATCH), Python (Pandas), Data Validation & Cleansing, Sales Analytics, Customer Segmentation, KPI Reporting, CRM Data Integration, Business Intelligence, Dashboard Design, Agile Collaboration, Stakeholder Engagement